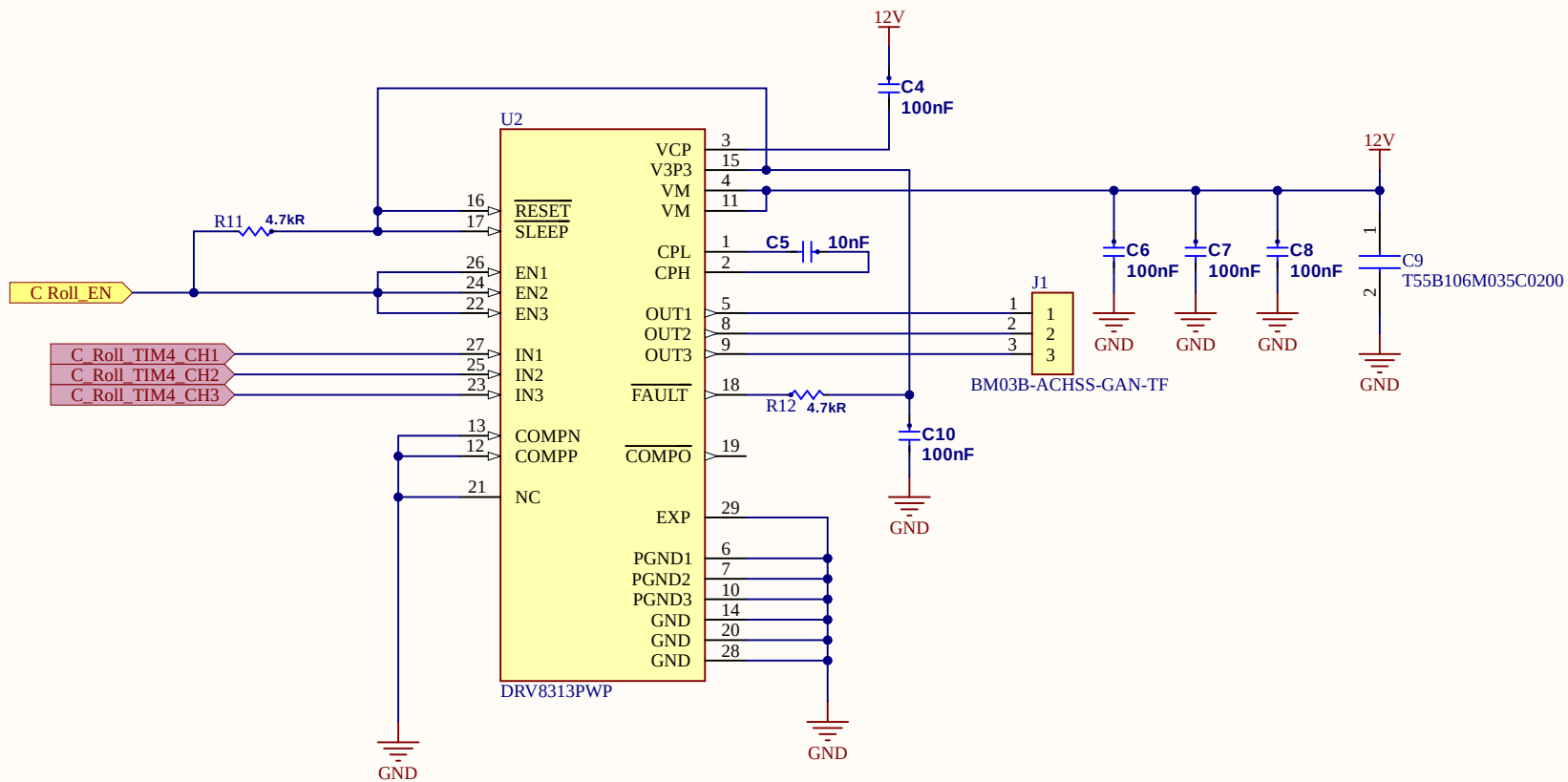
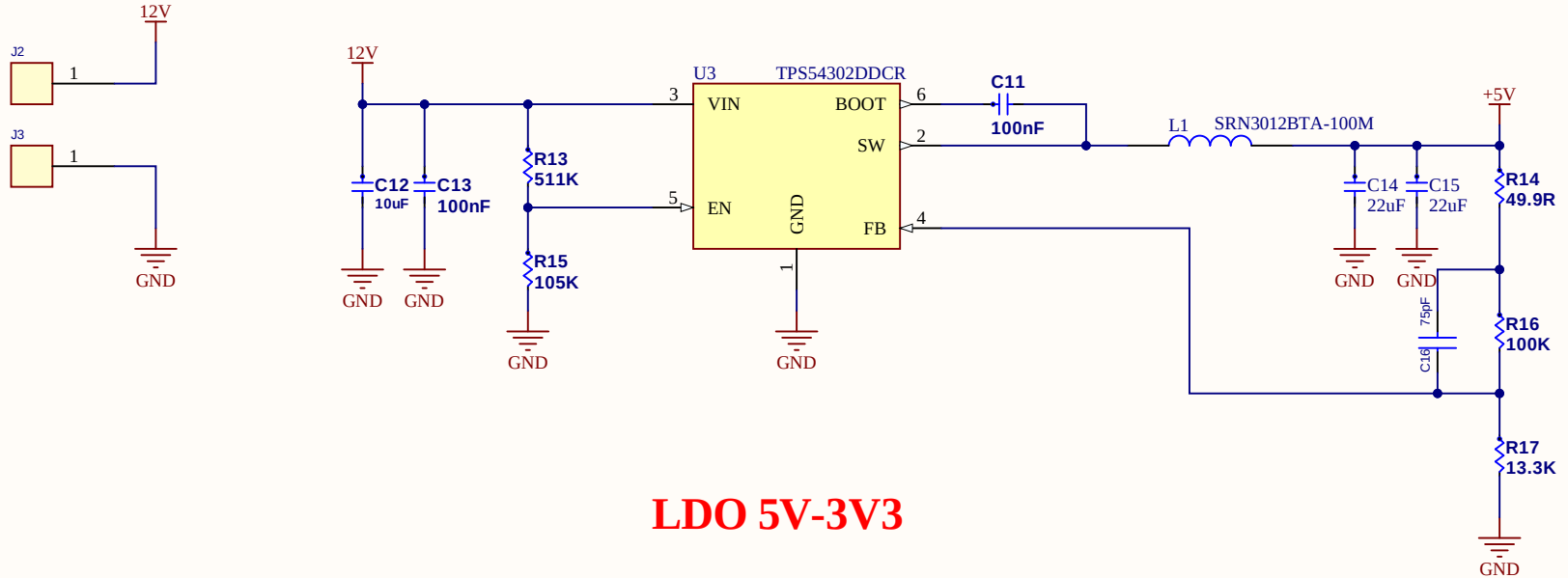


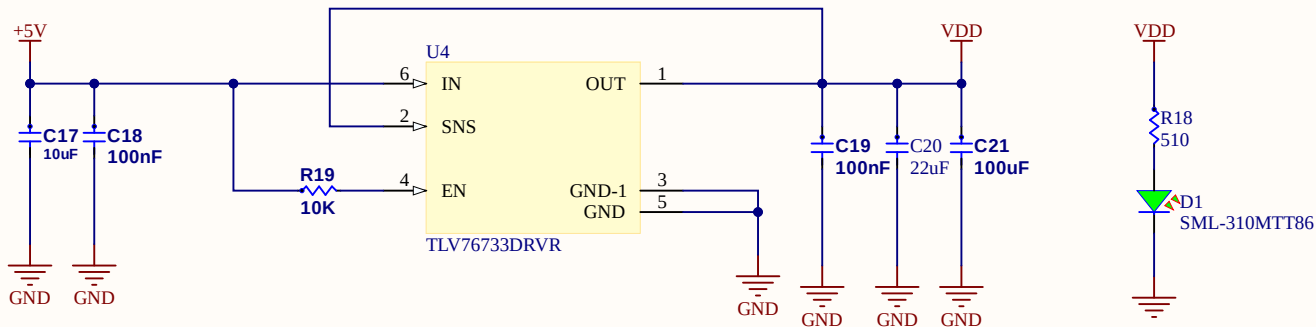


Title		
Size	Number	Revision
A		
Date:	3/24/2026	Sheet of
File:	F:\WORK\..\Driver.SchDoc	Drawn By:

Buck 12V - 5V



LDO 5V-3V3



Title		
Size	Number	Revision
A		
Date:	3/24/2026	Sheet of
File:	F:\WORK\..PWR.SchDoc	Drawn By:

12345678

AABBCCDD

C ROLL TO Main Rotor

P1	
1	MCU1 Main Left SWDIO
2	MCU1 Main Left SWCLK
3	MCU2 Main Left SWDIO
4	MCU2 Main Left SWCLK
5	MCU1 Main Right SWDIO
6	MCU1 Main Right SWCLK
7	MCU2 Main Right SWDIO
8	MCU2 Main Right SWCLK
9	ETH1 TX P
10	ETH1 TX N
11	ETH1 RX P
12	ETH1 RX N
13	RS232_ENC_Left_TX
14	RS232_ENC_Left_RX
15	RS232_ENC_Right_TX
16	RS232_ENC_Right_RX
17	PWM_1
18	ETH2 TX P
19	ETH2 TX N
20	ETH2 RX P
21	ETH2 RX N
22	RS232_ENC_Left_TX
23	RS232_ENC_Left_RX
24	RS232_ENC_Right_TX
25	RS232_ENC_Right_RX
26	PWM_2
27	SBUS_2
28	MX CAN L
29	MX CAN H
30	CAM_TRIGGER
31	C_Roll_SDA_IMU1
32	C_Roll_SCL_IMU1
33	C_Roll_SDA_MAG1
34	C_Roll_SCL_MAG1
35	RS232_TX1
36	RS232_RX1
37	RS232_TX2
38	RS232_RX2
39	RS232_TX3
40	RS232_RX3
41	RS232_TX4
42	RS232_RX4
43	RS232_TX5
44	RS232_RX5
45	RS232_TX6
46	RS232_RX6
47	RS232_TX7
48	RS232_RX7

C ROLL TO Arm Left

P2	
1	MCU1 Main Left SWDIO
2	MCU1 Main Left SWCLK
3	MCU2 Main Left SWDIO
4	MCU2 Main Left SWCLK
5	MCU1 Main Right SWDIO
6	MCU1 Main Right SWCLK
7	MCU2 Main Right SWDIO
8	MCU2 Main Right SWCLK
9	ETH1 TX P
10	ETH1 TX N
11	ETH1 RX P
12	ETH1 RX N
13	RS232_ENC_Left_TX
14	RS232_ENC_Left_RX
15	RS232_ENC_Right_TX
16	RS232_ENC_Right_RX
17	PWM_1
18	SBUS_1
19	RS232_TX1
20	RS232_RX1
21	RS232_TX2
22	RS232_RX2
23	RS232_TX3
24	RS232_RX3
25	RS232_TX4
26	RS232_RX4
27	MX CAN L
28	MX CAN H

C ROLL TO Arm Right

P3	
1	MCU1 Main Right SWDIO
2	MCU1 Main Right SWCLK
3	MCU2 Main Right SWDIO
4	MCU2 Main Right SWCLK
5	MCU1 Main Left SWDIO
6	MCU1 Main Left SWCLK
7	MCU2 Main Left SWDIO
8	MCU2 Main Left SWCLK
9	ETH2 TX P
10	ETH2 TX N
11	ETH2 RX P
12	ETH2 RX N
13	RS232_ENC_Left_TX
14	RS232_ENC_Left_RX
15	RS232_ENC_Right_TX
16	RS232_ENC_Right_RX
17	PWM_2
18	SBUS_2
19	RS232_TX2
20	RS232_RX2
21	RS232_TX3
22	RS232_RX3
23	RS232_TX4
24	RS232_RX4
25	RS232_TX5
26	RS232_RX5
27	MX CAN L
28	MX CAN H

Title		
Size	Number	Revision
A2	1/24/2025	
Date:	FWWORKS_ConnectorsSchDoc	Sheet of
File:		Drawn By:

12345678

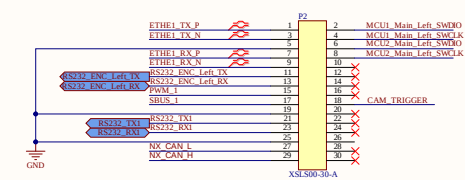


Figure 1: Pin connections for the MCLG Main Board. The diagram shows a 28-pin connector on the left and a 28-pin connector on the right. The left connector pins are: 1 (ETH2_TX_P), 2 (ETH2_TX_N), 3 (ETH2_RX_P), 4 (ETH2_RX_N), 5 (RS232_CIN_RxTx), 6 (RS232_CIN_RxTx), 7 (PDM_2), 8 (SDVS_2), 9 (RS232_RXQ), 10 (RS232_RXQ), 11 (NC_CAN_L), 12 (NC_CAN_L), 13 (NC_CAN_R), 14 (NC_CAN_R), 15 (GND), 16 (GND), 17 (GND), 18 (GND), 19 (GND), 20 (GND), 21 (GND), 22 (GND), 23 (GND), 24 (GND), 25 (GND), 26 (GND), 27 (GND), 28 (GND). The right connector pins are: 1 (MCLG_Main_RxTx_SWIO0), 2 (MCLG_Main_RxTx_SWIO1), 3 (MCLG_Main_RxTx_SWIO2), 4 (MCLG_Main_RxTx_SWIO3), 5 (MCLG_Main_RxTx_SWIO4), 6 (MCLG_Main_RxTx_SWIO5), 7 (MCLG_Main_RxTx_SWIO6), 8 (MCLG_Main_RxTx_SWIO7), 9 (MCLG_Main_RxTx_SWIO8), 10 (MCLG_Main_RxTx_SWIO9), 11 (MCLG_Main_RxTx_SWIO10), 12 (MCLG_Main_RxTx_SWIO11), 13 (MCLG_Main_RxTx_SWIO12), 14 (MCLG_Main_RxTx_SWIO13), 15 (MCLG_Main_RxTx_SWIO14), 16 (MCLG_Main_RxTx_SWIO15), 17 (MCLG_Main_RxTx_SWIO16), 18 (MCLG_Main_RxTx_SWIO17), 19 (MCLG_Main_RxTx_SWIO18), 20 (MCLG_Main_RxTx_SWIO19), 21 (MCLG_Main_RxTx_SWIO20), 22 (MCLG_Main_RxTx_SWIO21), 23 (MCLG_Main_RxTx_SWIO22), 24 (MCLG_Main_RxTx_SWIO23), 25 (MCLG_Main_RxTx_SWIO24), 26 (MCLG_Main_RxTx_SWIO25), 27 (MCLG_Main_RxTx_SWIO26), 28 (MCLG_Main_RxTx_SWIO27).

Title			
Size A2	Number		Revision
Date: File:	3/24/2026 F:\WORK\...ConnectorSchDoc	Sheet of Drawn By:	

Comment	Description	Designator	Footprint	LibRef	Quantity
C0603C104M5RACTU	CAP CER 0.1UF 50V X7R 0603	C1, C2, C4, C6, C7, C8, C10, C22	CAP0603-M	CMP-00691-1	8
CL21A476MQYNNNE	CAP CER 47UF 6.3V X5R 0805	C3, C23	CAP0805-M	885012107014	2
C0603C103M5RACTU	CAP CER 10NF 50V X7R 0603	C5	CAP0603-M	CMP-00691-1	1
T55B106M035C0200	Capacitor	C9	CAPC3528X210N	T55B106M035C0200	1
CC0402KRX5R9BB104	CAP CER 0.1UF 50V X5R 0402	C11, C13, C18, C19	CAP0402-M	CMP-00265-1	4
GRM188R6YA106MA7 3D	CAP CER 10UF 35V X5R 0603	C12, C17	CAP0603-M	CMP-01031-1	2
GRM158R61A226ME1 5D	CAP CER 22UF 10V X5R 0402	C14, C15, C20	CAP0402-M	CMP-00520-1	3
75pF	CAP CER 75PF 25V C0G/NP0 0402	C16	C_0402	CAP_Dup1_16	1
CL21A107MQYNNWE	CAP CER 100UF 6.3V X5R 0805	C21	CAP0805-M	CMP-00446-1	1
SML-310MTT86	LED GREEN CLEAR 0603 SMD	D1	SML-310MTT86- Footprint-1	CMP-01130-2	1
BM03B-ACHSS-GAN- TF	ACH Series 3 Position 1.2 mm Pitch Disconnectable Crimp Style Housing	J1	JST_BM03B-ACHSS- GAN-TF	BM03B-ACHSS-GAN- TF	1
CON1	SOLDER PADS	J2, J3	wire1_220s	CON1	2
SRN3012BTA-100M	10uH	L1	SRN3012BTA330M	SRN3012BTA-330M	1
XLS00-52-B		P1	XLS00-52-B	XLS00-52-B	1
XLS00-30-A		P2, P3	XLS00-30-A	XLS00-30-A	2
BSS138	N-Channel 50 V 220mA (Ta) 350mW (Ta) Surface Mount SOT-23-3	Q1, Q2, Q3, Q4	SOT95P240X111-3N	BSS138	4
RC0603FR-072K2L	RES SMD 2.2K OHM 1% 1/10W 0603	R1, R2, R20, R21	RES0603	CMP-01404-1	4
RC0603JR-074K7L	RES SMD 4.7K OHM 5% 1/10W 0603	R3, R4, R11, R12, R23	RES0603	CMP-01404-1	5
2.2k	Res Thick Film 0603 2.2K Ohm 1% 0.1W(1/10W) +/- 100ppm/C Epoxy Pad SMD T/R	R5, R7, R8, R10, R24, R26, R27, R29	RES0603	RC0603FR-072K2L	8
MCR03EZPJ100	RES SMD 10 OHM 5% 1/10W 0603	R6, R9, R25, R28	RES0603	CMP-01078-4	4
511K	RES SMD 511K OHM 1% 1/16W 0402	R13	RES0402	CMP-00299-1_1	1
49.9R	RES SMD 49.9R OHM 1% 1/16W 0402	R14	RES0402	CMP-00299-1_1	1
105K	RES SMD 105K OHM 1% 1/16W 0402	R15	RES0402	CMP-00299-1_1	1
100K	RES SMD 100K OHM 1% 1/16W 0402	R16	RES0402	CMP-00299-1_1	1
13.3K	RES SMD 13.3K OHM 1% 1/16W 0402	R17	RES0402	CMP-00299-1_1	1
RT0402FRE07510RL	RES SMD 510 OHM 1% 1/16W 0402	R18	RES0402	RC0402FR-0722KL	1
CRCW040210K0JNED	RES SMD 10K OHM 5% 1/16W 0402	R19	RES0402	CMP-00034-1	1
RC0603FR-071KL	RES SMD 1K OHM 1% 1/10W 0603	R22	RES0603	CMP-00067-1	1
AS5048B-HTSP-500	No Description Available	U1	TRANS_HTSP- 500_AMS	AS5048B-HTSP-500	1
DRV8313PWP	65-V max, 3-A peak 3- phase motor driver	U2	SOP65P640X120-29N	DRV8313PWP	1
TPS54302DDCR	4.5-V to 28-V Input, 3- A Output, EMI Friendly Synchronous Step-Down Converter	U3	SOT95P280X110-6N	TPS54302DDCR	1
TLV76733DRVR	Adjustable- and fixed output, 1-A, 16-V, positive-voltage low- dropout (LDO) linear regulator	U4	SON65P200X200X80- 7N	TLV76733DRVR	1
ICM-42688-P	Accelerometer, Gyroscope, 6 Axis Sensor Output	U5	PQFN50P300X250X97- 14N	ICM-42688-P	1